

provide a hardware keyboard as well, and since some users still prefer these physical keyboards to onscreen virtual keyboards, that may be an interesting option.

- ▶ **On-screen resolution:** Windows Phone supports two screen resolutions, 800x480 (WVGA) and 480x320 (HVGA). The former is more common and more appropriate for what will likely prove to be the most popular Windows Phone configuration. But hardware makers are free to use the lower-resolution screen type, and will likely do so in smaller devices, including those with slide-out hardware keyboards.
- ▶ **Accelerometer:** First brought to smartphones by the iPhone, an accelerometer is a hardware component that lets the phone detect the amount of acceleration along all the six different axes. What this results in is the ability of the Windows Phone to be able to detect and act upon the device being tilted in any particular direction. The accelerometer most primarily can be seen being used when you rotate the device horizontally and the display rotates to accommodate the new orientation. Similarly for a racing game, moving the screen left to right as you play can help you steer the car.
- ▶ **Assisted GPS (A-GPS):** All the Windows Phones come with this latest GPS (global positioning system) hardware, providing quicker startup and better accuracy, the latter of which is key to a US-based requirement that will allow 911 dispatchers to find smartphone users in an emergency.
- ▶ **Compass:** Windows Phones ship with an internal compass, which works in concert with the GPS and other location sensors (including Wi-Fi and cellular connection) to accurately find your location and supply information about the direction you're facing. As originally delivered in late 2010, the compass hardware in Windows Phone works only with the built-in Bing Maps functionality. Microsoft will provide programming libraries to developers to access the compass later, however.
- ▶ **Light sensor:** Thanks to the built-in light sensor, the Windows Phone camera can accurately gauge illumination requirements for the flash and produce accurate and clear low-light photos.